

Companion XXXIII: Assembly Bias in the Relaxation-Driven Cyclic Cosmology

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Abstract

Assembly bias describes the phenomenon that halos of identical mass cluster differently depending on their formation history. In the standard cosmological model, this effect emerges from the interplay between large-scale tidal fields, merger histories, and the stochasticity of hierarchical growth. In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally altered. The relaxation-modified potential evolution introduces a temporal memory into the growth of structure, which reshapes the relation between halo formation time, environment, and clustering strength. This Companion develops a continuous, non-discrete description of assembly bias in RDCC, showing how the relaxation scale k_{rel} imprints itself on the temporal architecture of halo growth and on the spatial coherence of the cosmic web.

1 Introduction

The concept of assembly bias challenges the simplest picture of structure formation. If halo mass were the only relevant variable, halos of identical mass would exhibit identical clustering properties. Yet numerical simulations and observations reveal a more intricate reality: halos that form earlier cluster more strongly, even when their masses are the same. This sensitivity to formation history indicates that the Universe retains a memory of the temporal sequence of collapse events.

In RDCC, this memory is not an emergent byproduct of hierarchical growth but a direct consequence of the relaxation-modified potential evolution. The suppression of small-scale growth and the enhancement of large-scale coherence introduce a temporal structure into the gravitational field itself. As a result, the formation time of a halo becomes intertwined with the relaxation scale, and assembly bias acquires a distinctive, theoretically predictable form.

2 Temporal Memory in RDCC Structure Formation

In the standard cosmological model, the gravitational potential evolves primarily through the linear growth factor, and its time dependence is smooth and monotonic. RDCC modifies this picture by introducing a relaxation-driven component that depends on scale. Modes with $k \gg k_{\text{rel}}$ experience a reduced potential decay, while modes with $k \ll k_{\text{rel}}$ retain a more coherent temporal evolution.

This scale-dependent behavior means that the gravitational field carries a memory of its past configuration. When a halo begins to collapse, the potential it experiences is not solely determined by its local density contrast but also by the relaxation history of the surrounding modes. Halos forming in regions dominated by large-scale coherence inherit a smoother collapse trajectory,

while halos forming in regions where small-scale modes are suppressed experience a delayed or softened collapse.

This temporal memory is the foundation of RDCC assembly bias.

3 Formation Time and Environmental Dependence

In RDCC, the formation time of a halo is not an isolated parameter but a reflection of its environment across multiple scales. Halos embedded in regions where the potential evolves coherently tend to form earlier, even if their local density contrasts are modest. Conversely, halos in environments dominated by suppressed small-scale modes form later, as the relaxation mechanism slows the growth of perturbations below k_{rel} .

This interplay produces a continuous spectrum of formation histories. Early-forming halos emerge from regions where the relaxation scale reinforces the coherence of the surrounding potential, while late-forming halos arise in environments where the relaxation mechanism inhibits the collapse of small-scale fluctuations. The result is a natural, scale-dependent modulation of halo formation times that directly feeds into assembly bias.

4 Clustering Signatures of RDCC Assembly Bias

The clustering of halos reflects the coherence of the gravitational field in which they formed. In RDCC, early-forming halos inherit the imprint of large-scale relaxation, which enhances their spatial coherence. Their clustering amplitude is therefore elevated relative to halos of the same mass that formed later. This effect is not a discrete bifurcation but a smooth gradient across formation times.

Late-forming halos, shaped by environments where small-scale modes were suppressed, exhibit weaker clustering. Their spatial distribution reflects the diminished influence of coherent large-scale potentials during their collapse. The contrast between early and late formation is thus amplified by the relaxation scale, which acts as a temporal filter on the gravitational field.

The resulting assembly bias is a continuous, scale-dependent phenomenon that emerges naturally from the RDCC framework.

5 Implications for the Cosmic Web

Assembly bias in RDCC extends beyond individual halos and influences the morphology of the cosmic web. Regions dominated by early-forming halos tend to develop more pronounced filaments and stronger connectivity, reflecting the coherence of the underlying potential. In contrast, regions populated by late-forming halos exhibit a more diffuse structure, with weaker filamentary alignment and larger voids.

This environmental dependence creates a subtle but measurable modulation of the cosmic web. The relaxation scale determines the degree to which large-scale modes shape the web's architecture, while the suppression of small-scale modes influences the fragmentation of filaments and the depth of voids. The interplay between these effects provides a rich observational landscape for testing RDCC.

6 Conclusion

Assembly bias in the Relaxation-Driven Cyclic Cosmology arises from the temporal structure of the gravitational potential. The relaxation-modified evolution of perturbations introduces a memory into the collapse process, linking halo formation times to the coherence of their environments. This connection produces a continuous, scale-dependent form of assembly bias that reshapes halo clustering and the architecture of the cosmic web.

The results of this Companion establish assembly bias as a natural and predictive feature of RDCC. They also provide a conceptual bridge between the non-linear halo model of Companion XXXI and future studies of galaxy formation, cluster astrophysics, and cosmic web morphology.

References

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